

Project Planning

17-313: Foundations of Software Engineering

<https://cmu-313.github.io>

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Fall 2026

Gentle Reminder

- Please close laptops
- Tablets with stylus ok



Administrivia

- Problem Set 1 on Canvas, due September 2 @ 11:59pm
- Quiz on Thursday
- Reading and participation activity due ***before*** class on Thursday

Today's Learning Goals

- Recognize the importance of process
- Identify why software development has project characteristics
- Understand the elements of Scrum
- Create and evaluate user stories
- Use milestones for planning and progress measurement
- Understand the difficulty of measuring progress



How the Customer explained it



What the Project Manager understood



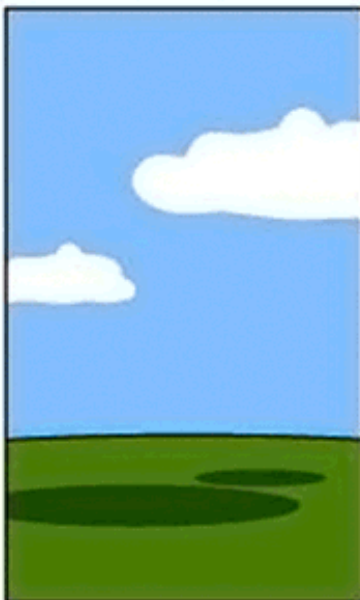
How the Analyst designed it



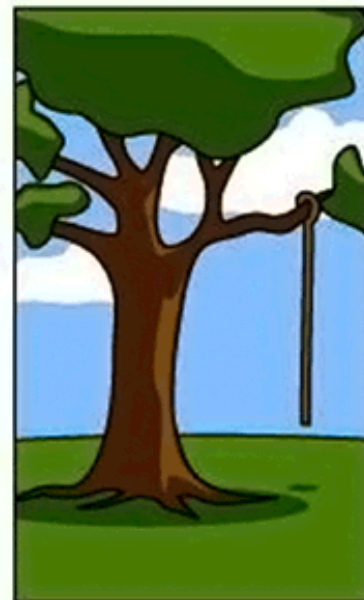
What the Programmer wrote



What the Business Consultant presented



How the Project was documented



What Operations installed



How the Customer was billed



How the Solution was supported



What the Customer really needed

Software Process

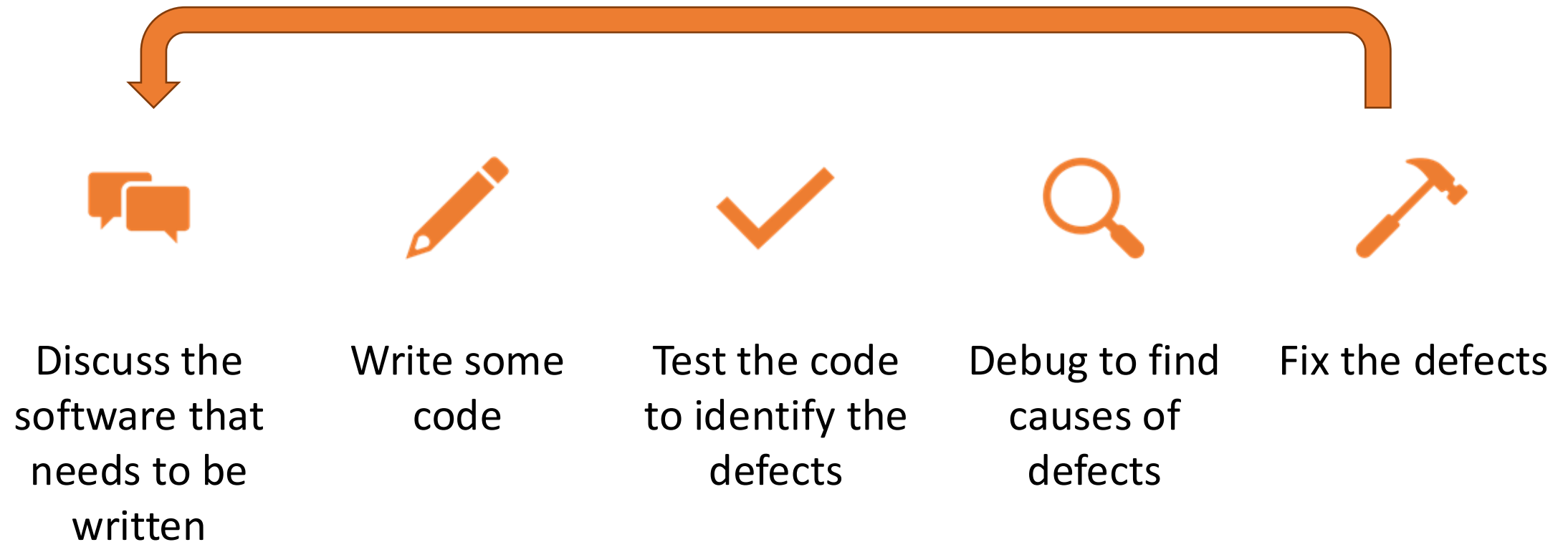
“The set of activities and associated results that produce a software product”

Sommerville, SE, ed. 8

What does this mean?
What else can we do apart
from coding?
Processes are key
concerns.

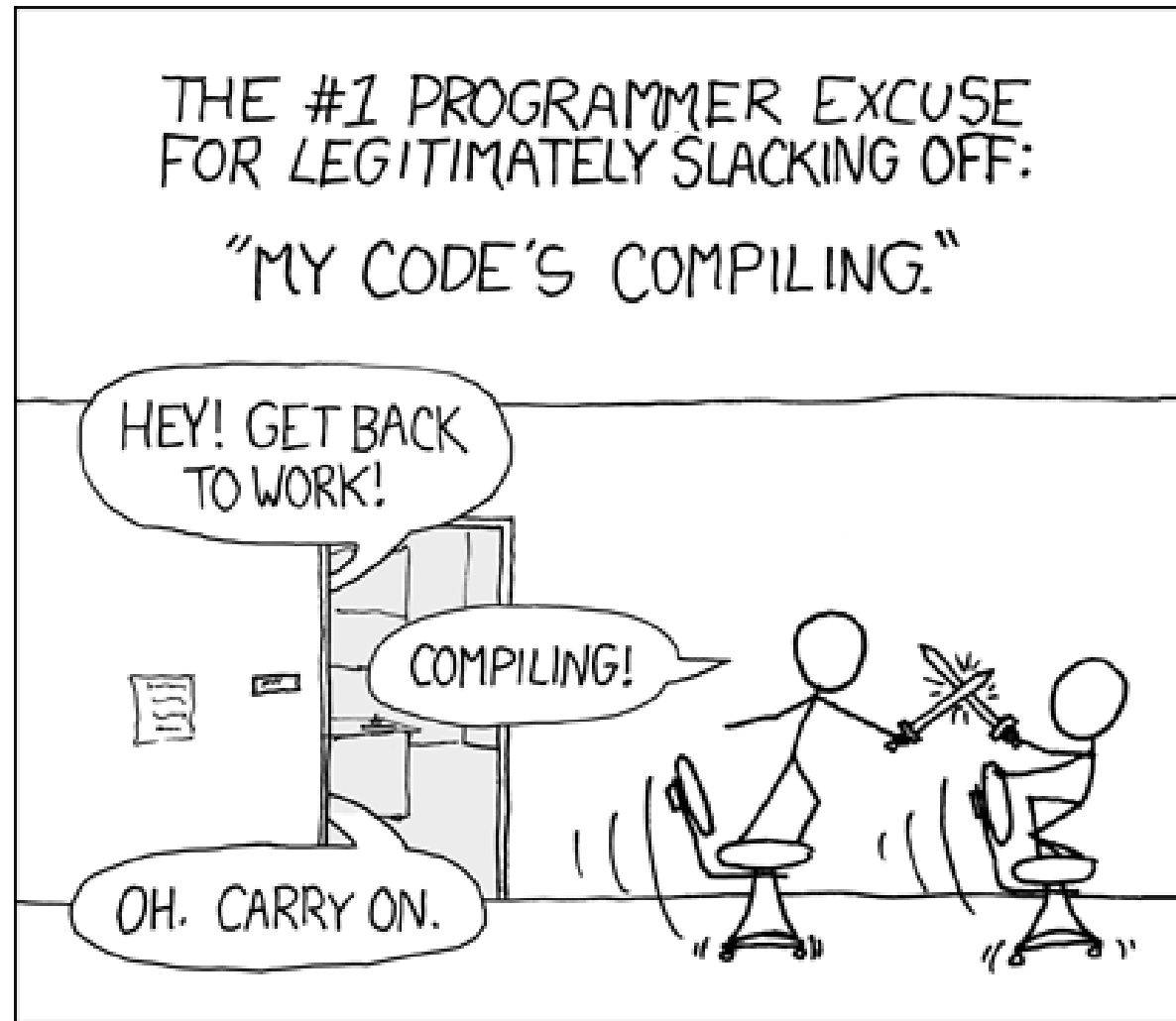
**Software Engineering Principles,
practices (technical and non-
technical) for confidently building
high-quality software.**

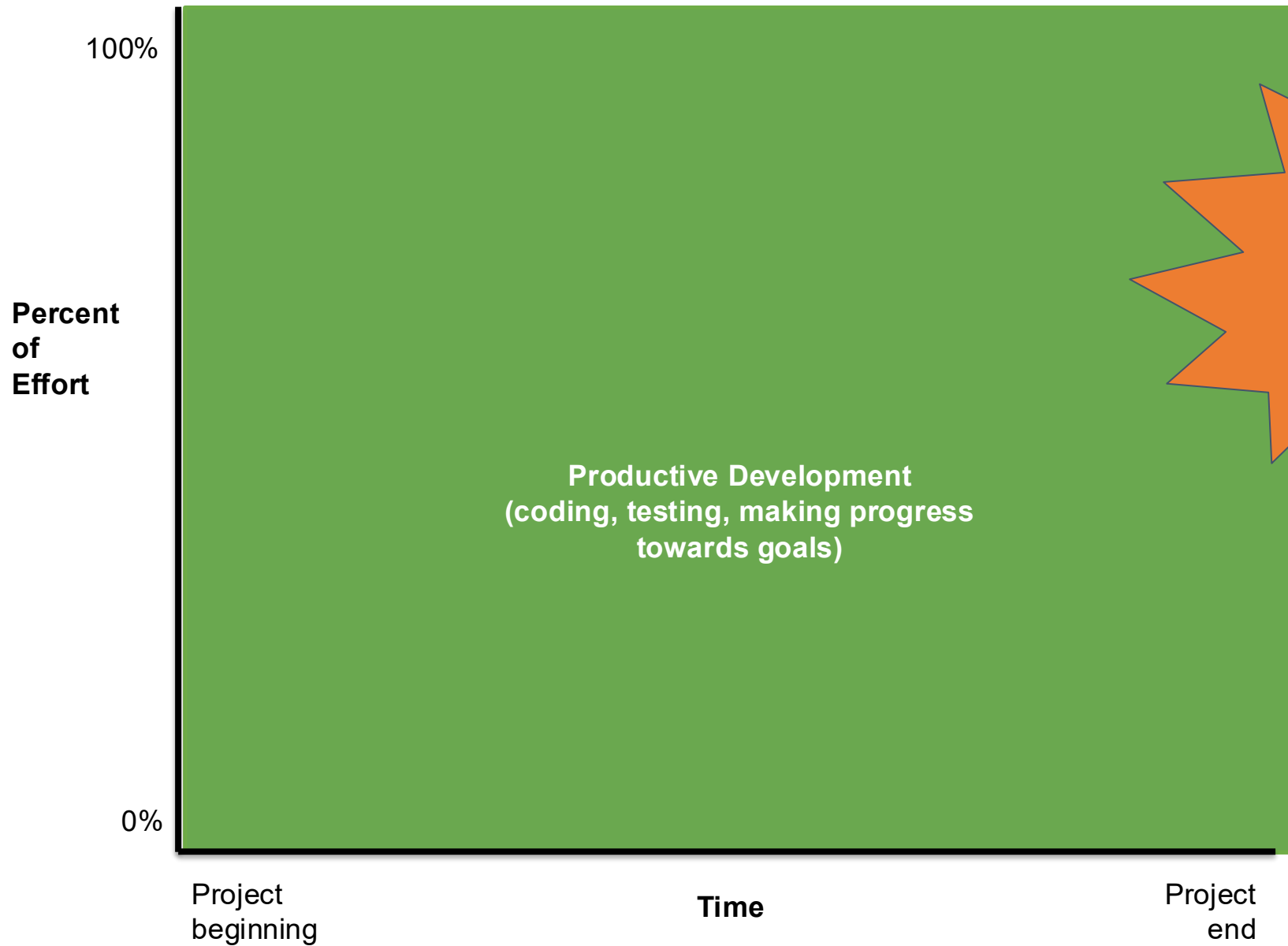
How to develop software???

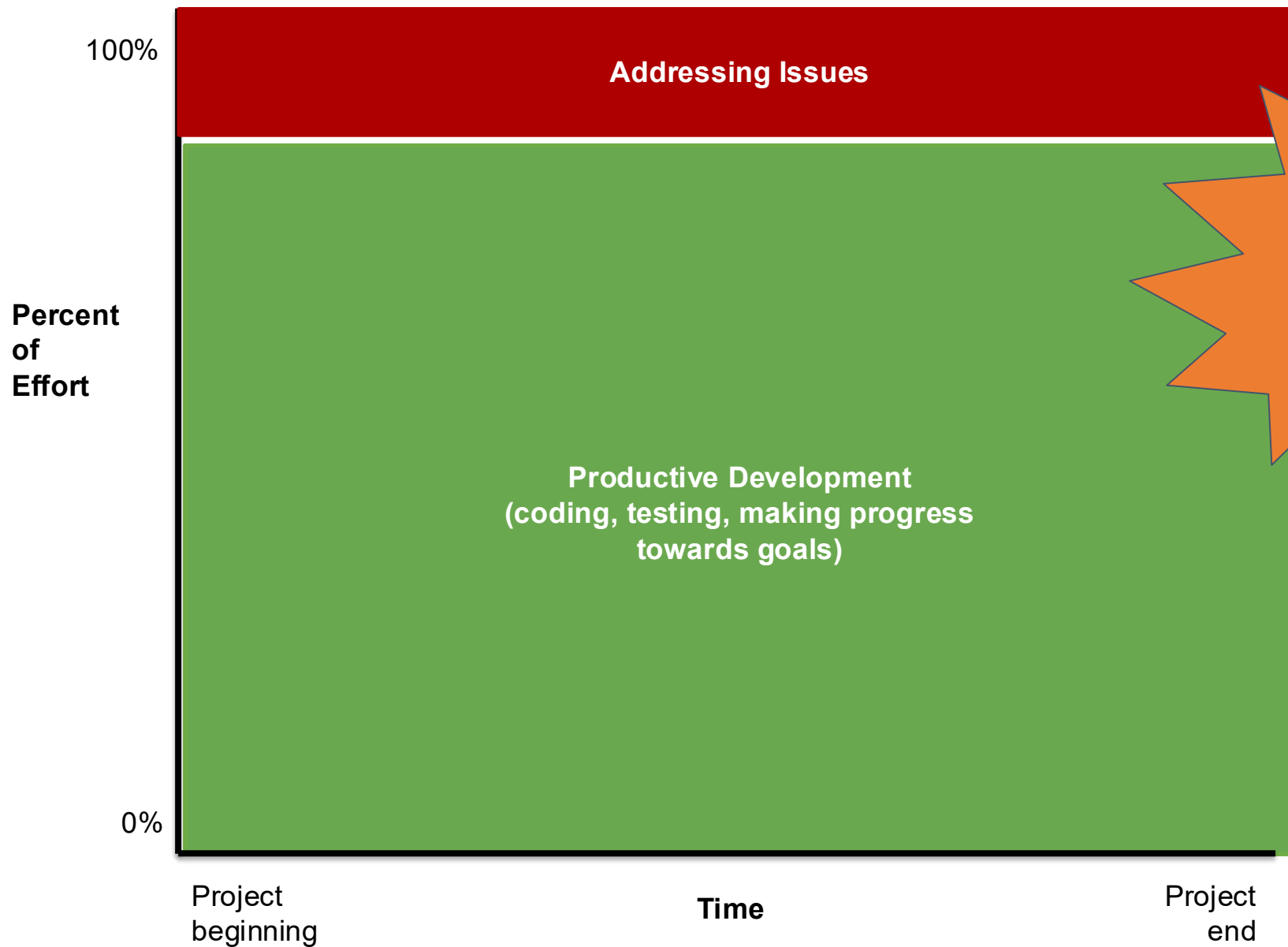


What does a software engineer's day look like?

- How many hours do they spend in meetings, coding, testing, debugging, etc.?

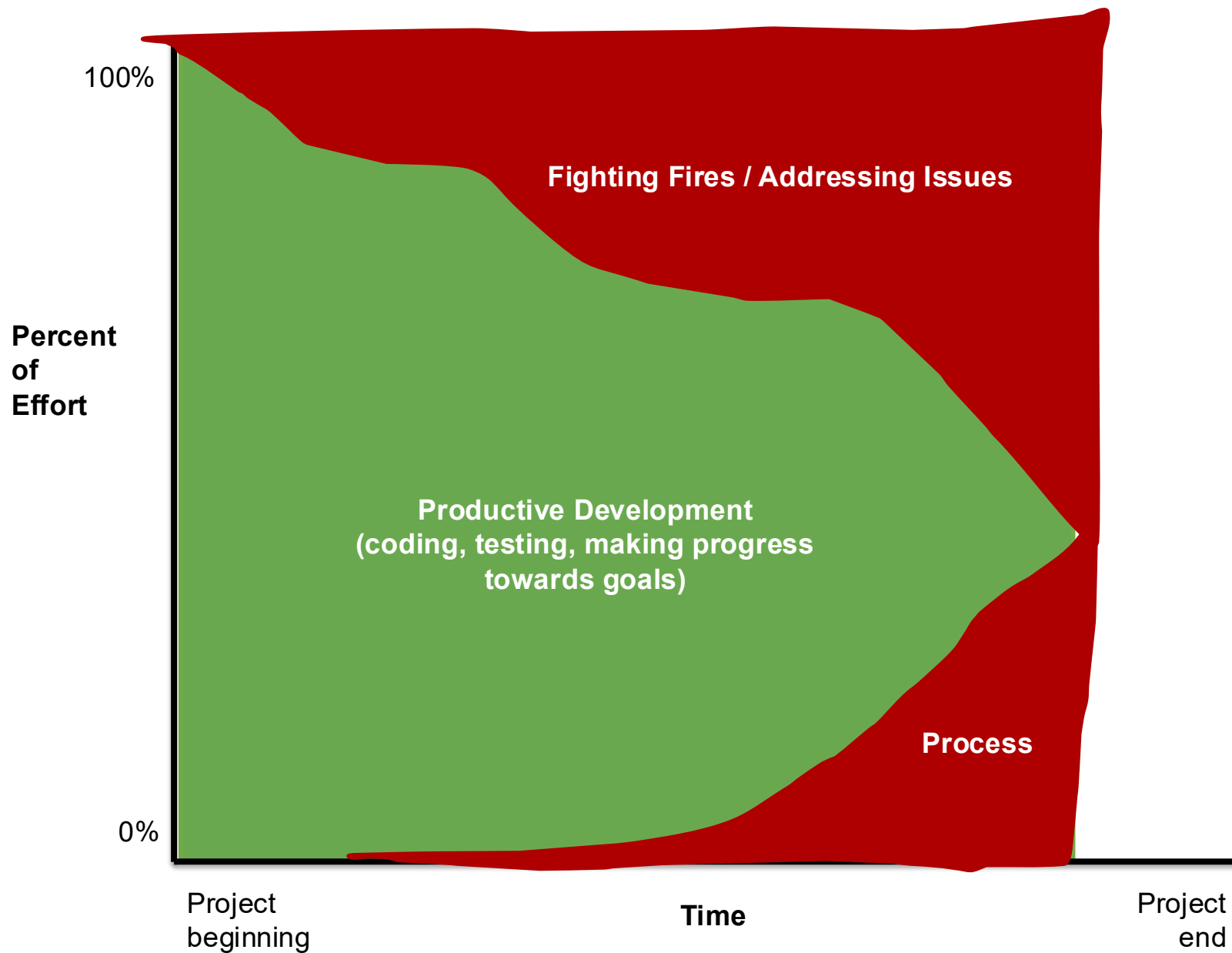






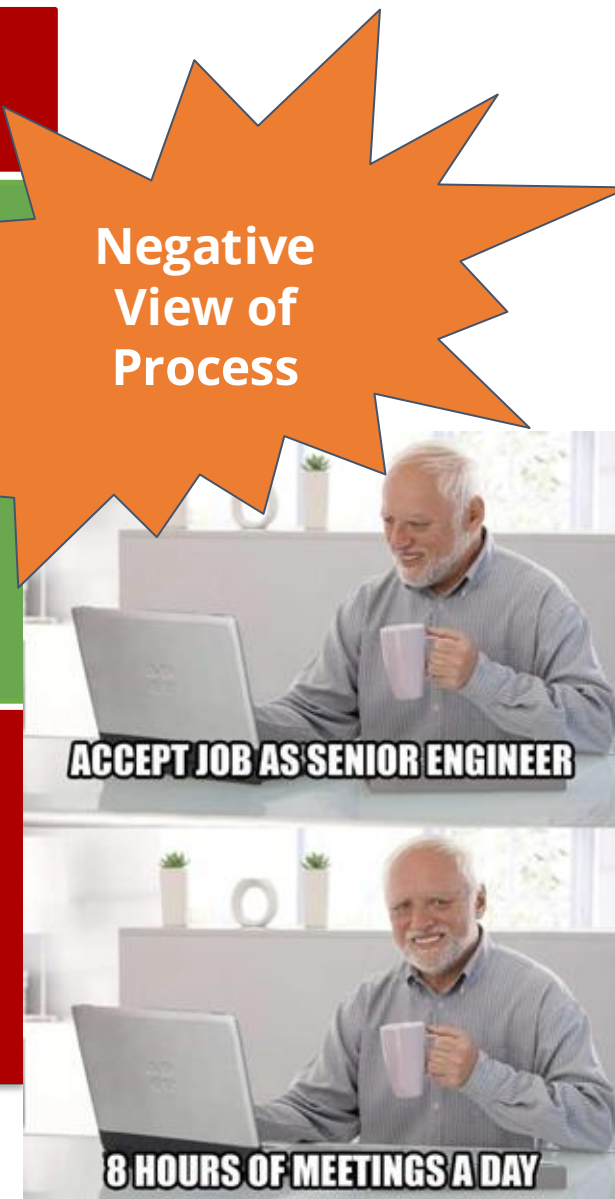
What happens when ...

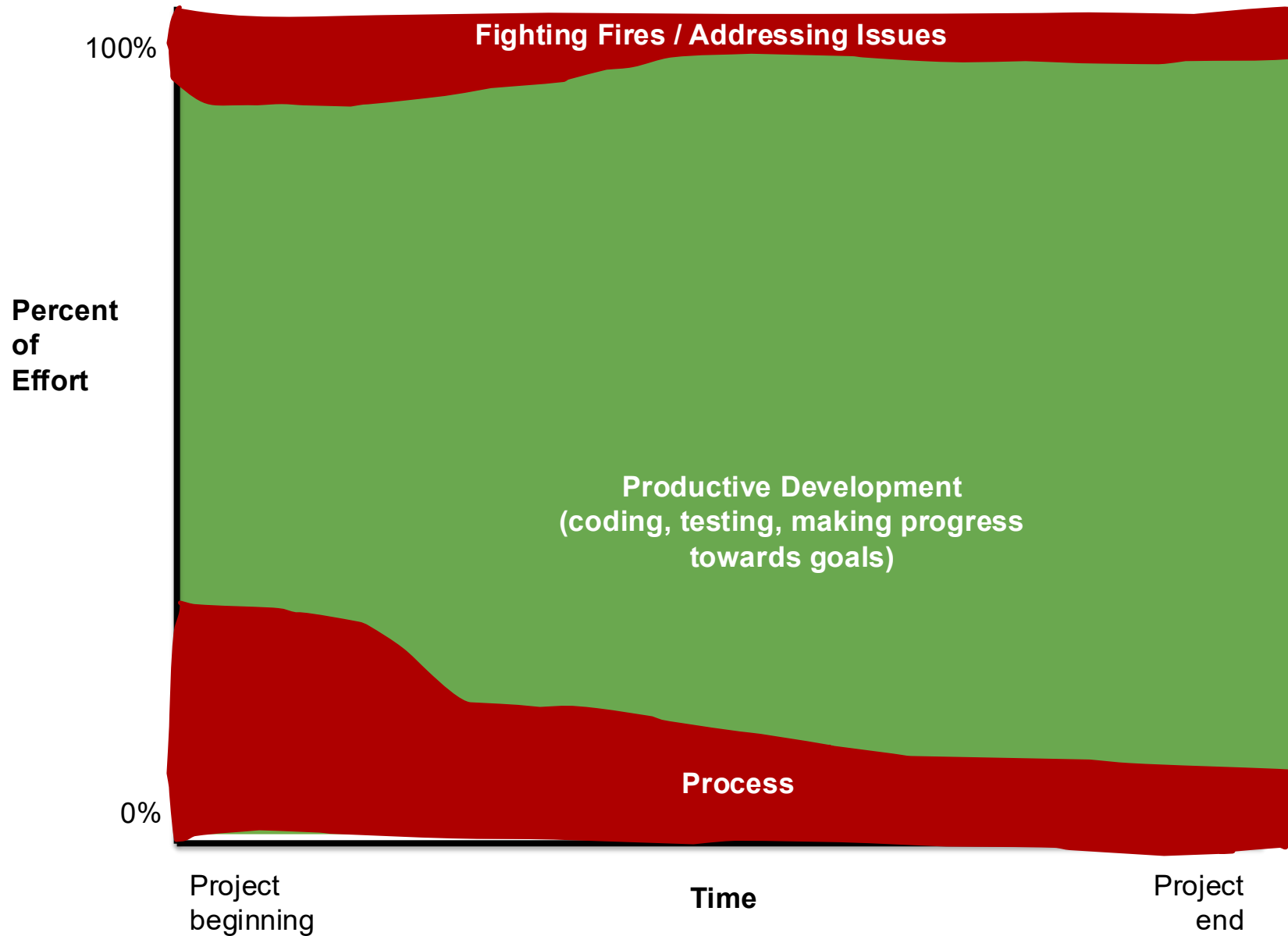
- **Uncontrolled Scope Creep:** Informal agreements balloon the project scope by 25-50%.
- **Late Failure:** Critical requirement and design flaws are discovered only during final testing.
- **Lost Defects:** Informal bug tracking (emails/hallway chats) leads to forgotten fixes.
- **Code Chaos:** Lack of version control leads to overwritten files and lost work.



Let's improve the reliability of this process

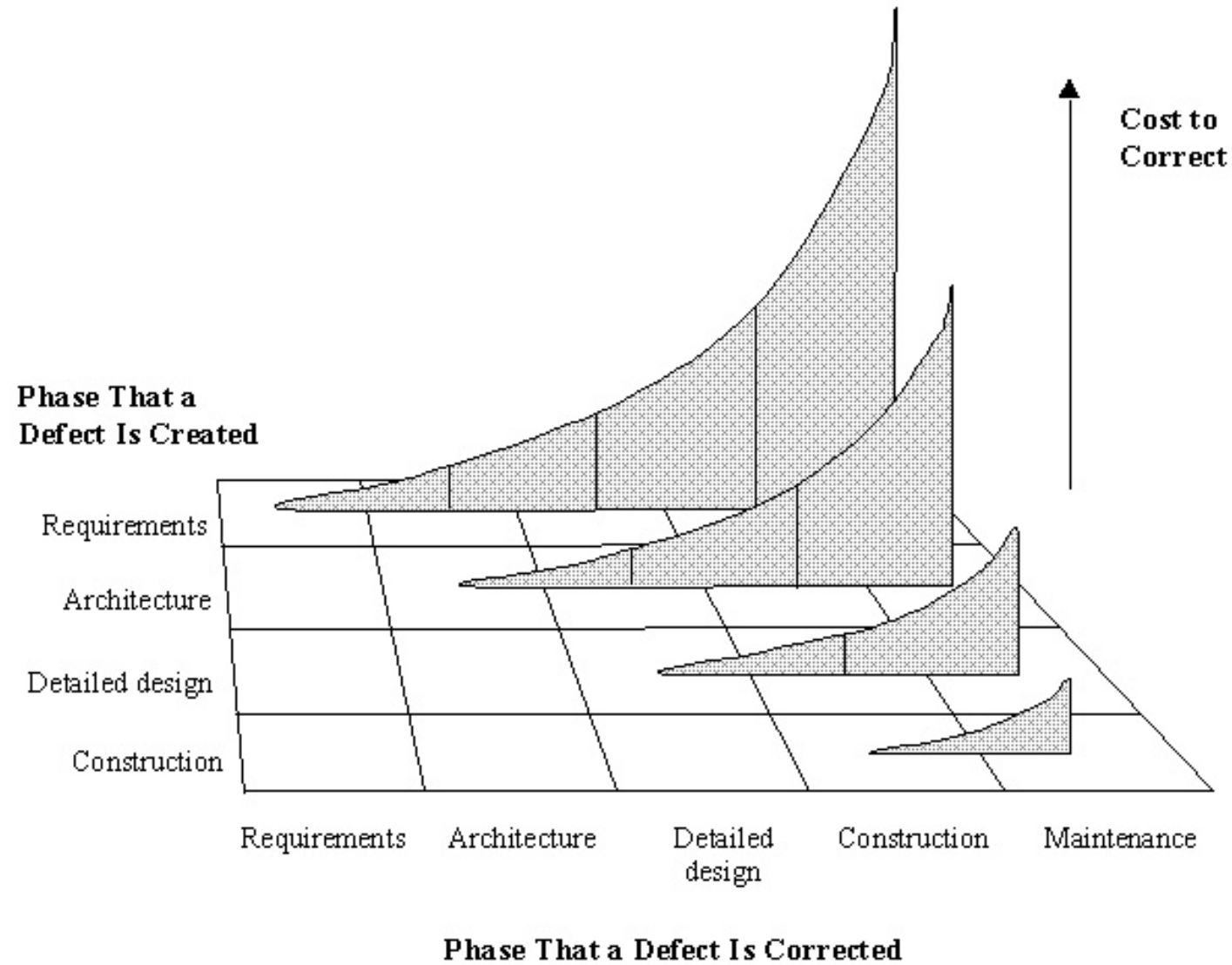
- Writing down all requirements
 - Review requirements
 - Require approval for all changes to requirements
- Use version control for all changes
 - Code Reviews
- Track all work items
 - Break down development into smaller tasks
 - Write down and monitor all reported bugs
 - Hold regular, frequent status meetings
- Plan and conduct quality assurance
- Employ a DevOps framework to push code between developers and operations





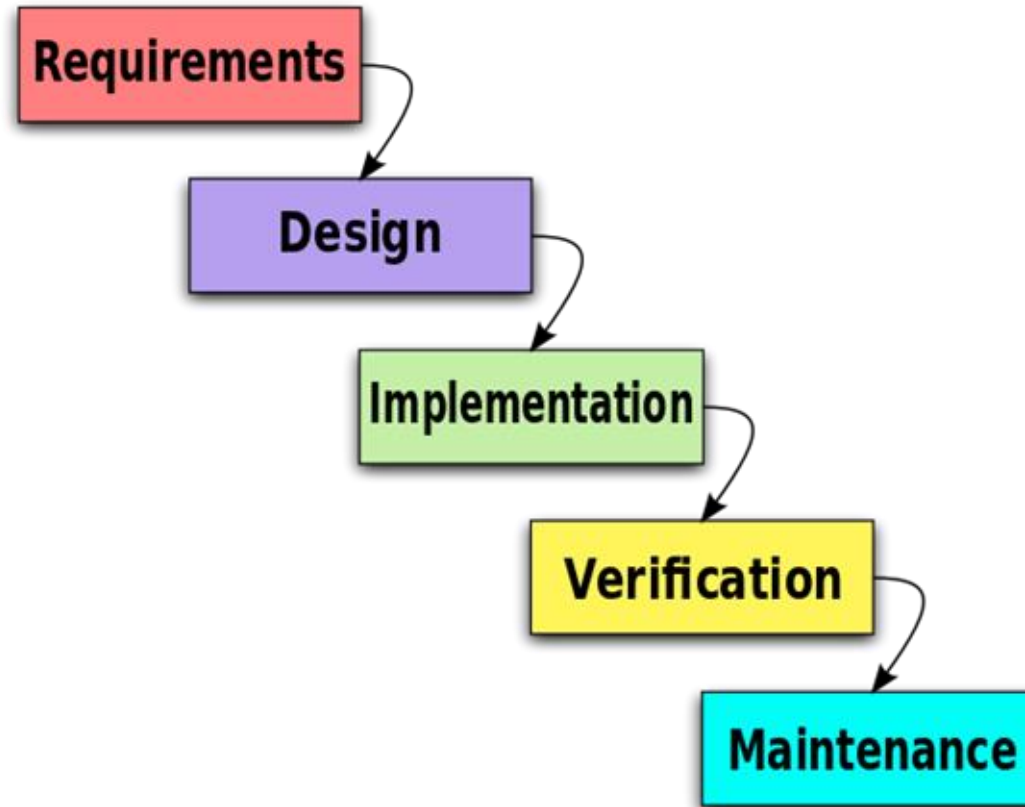
Hypothesis: Process increases flexibility and efficiency

Ideal Curve: Upfront investment for later greater returns



Copyright 1998 Steven C. McConnell. Reprinted with permission
from *Software Project Survival Guide* (Microsoft Press, 1998).

Waterfall model was the original software process

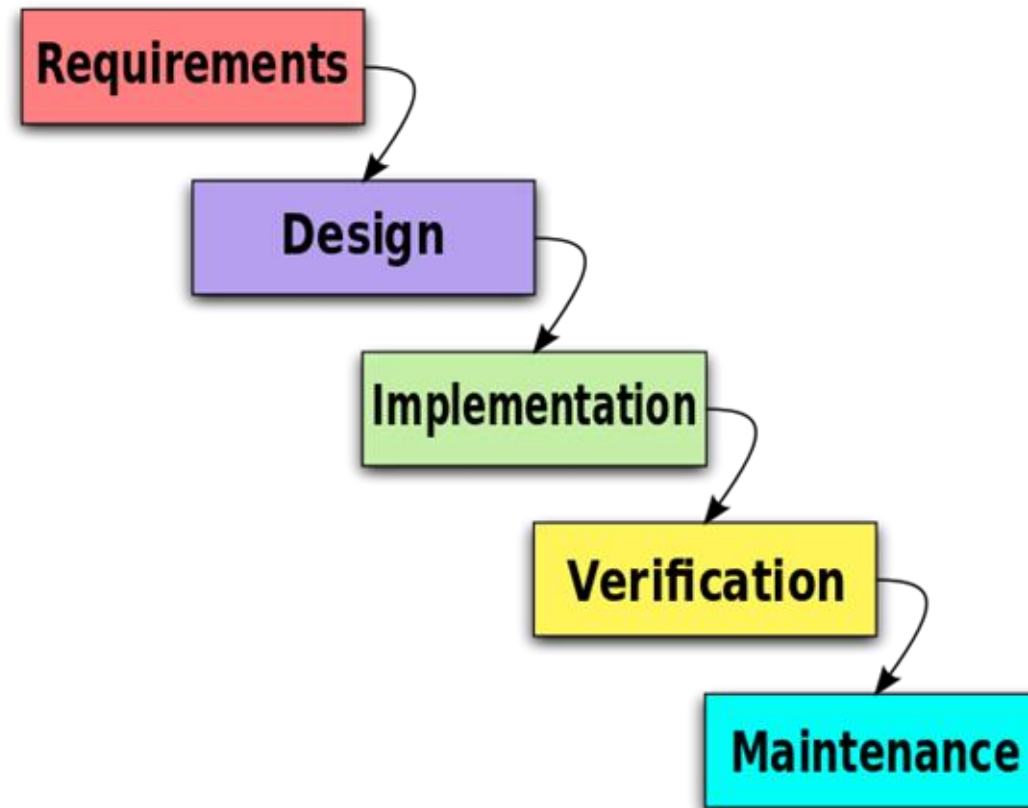


Waterfall diagram CC-BY 3.0 [Paulsmith99](#) at [en.wikipedia](#)

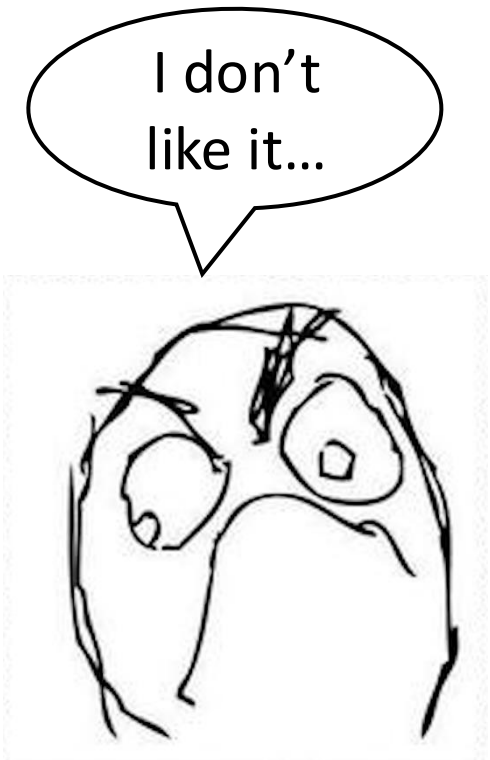
... akin to processes pioneered in mass manufacturing (e.g., by Ford)



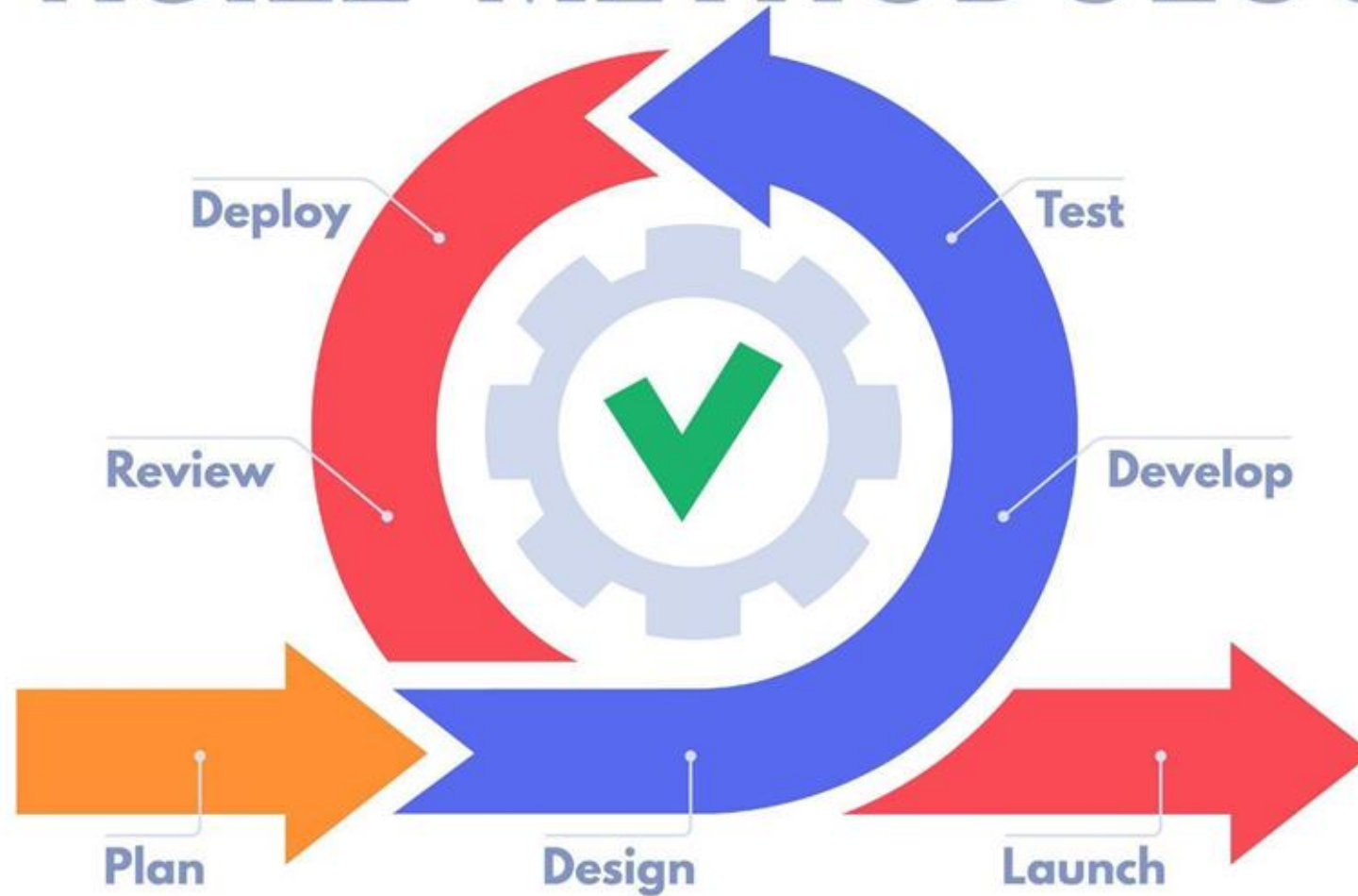
What could go wrong?



Waterfall diagram CC-BY 3.0 [Paulsmith99](#) at [en.wikipedia](#)



AGILE METHODOLOGY



Agile manifesto

“We are uncovering better ways of developing software by doing it and helping others do it. Through this work we have come to value:

<https://agilemanifesto.org/>

Individuals and interactions over processes and tools
Working software over comprehensive documentation
Customer collaboration over contract negotiation
Responding to change over following a plan

That is, while there is value in the items on the right, we value the items on the left more.”

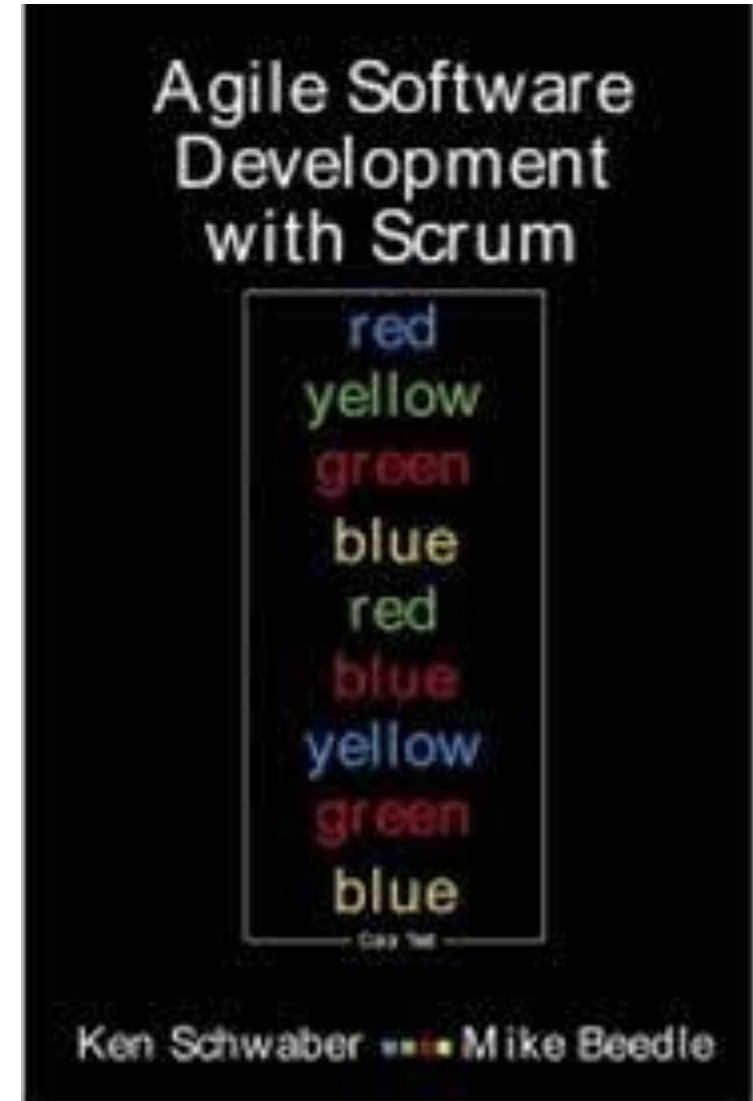
Agile manifesto

Twelve high-level principles, examples include:

- *“Our highest priority is to satisfy the customer through early and continuous delivery of valuable software.”*
- *“Working software is the primary measure of progress”*
- *“Continuous attention to technical excellence and good design enhances agility.”*
- *“Simplicity—the art of maximizing the amount of work not done—is essential.”*

Scrum

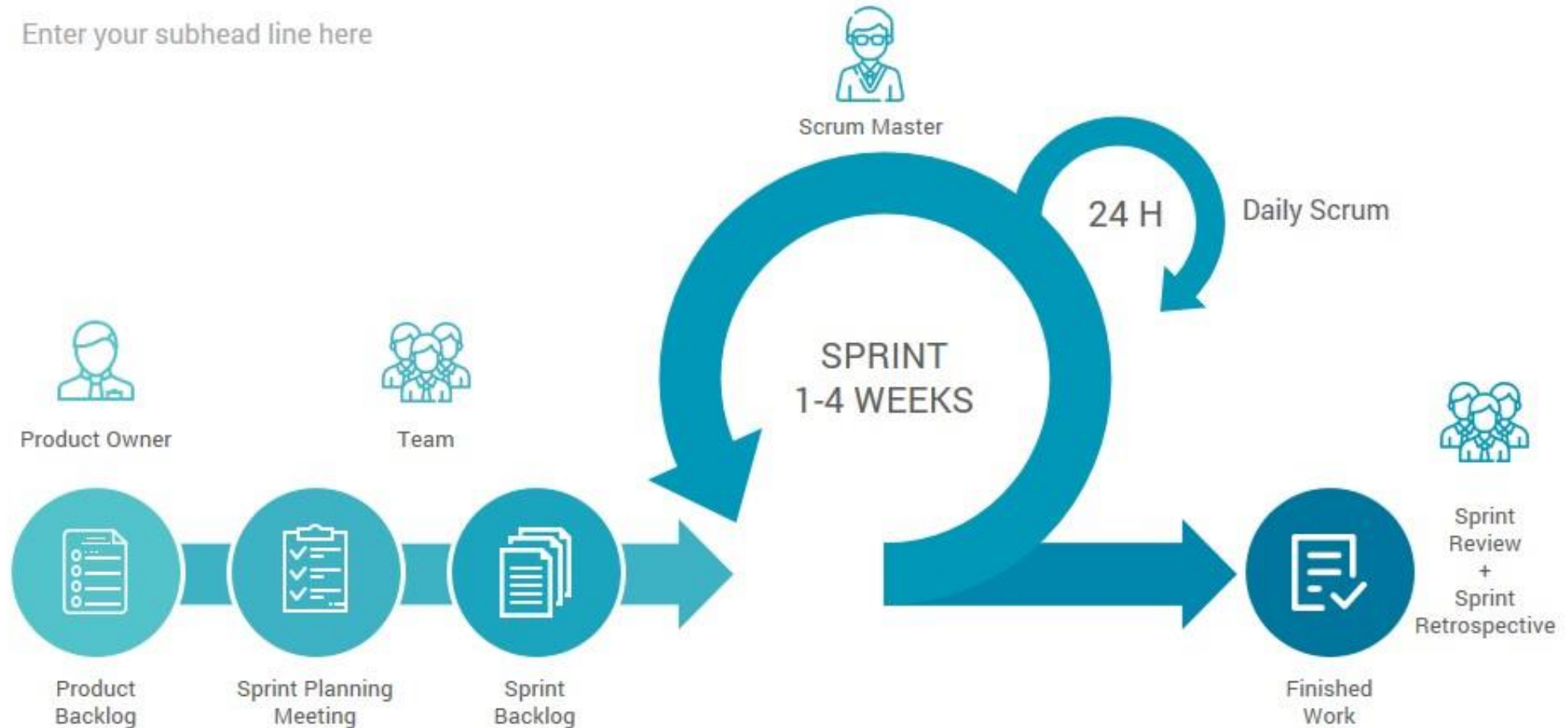
(Only a brief intro)



Elements of Scrum

Scrum Process

Enter your subhead line here



Backlogs

The **product backlog** is all the features for the product

The **sprint backlog** is all the features that will be worked on for that sprint. These should be broken down into discrete tasks:

- Fine-grained

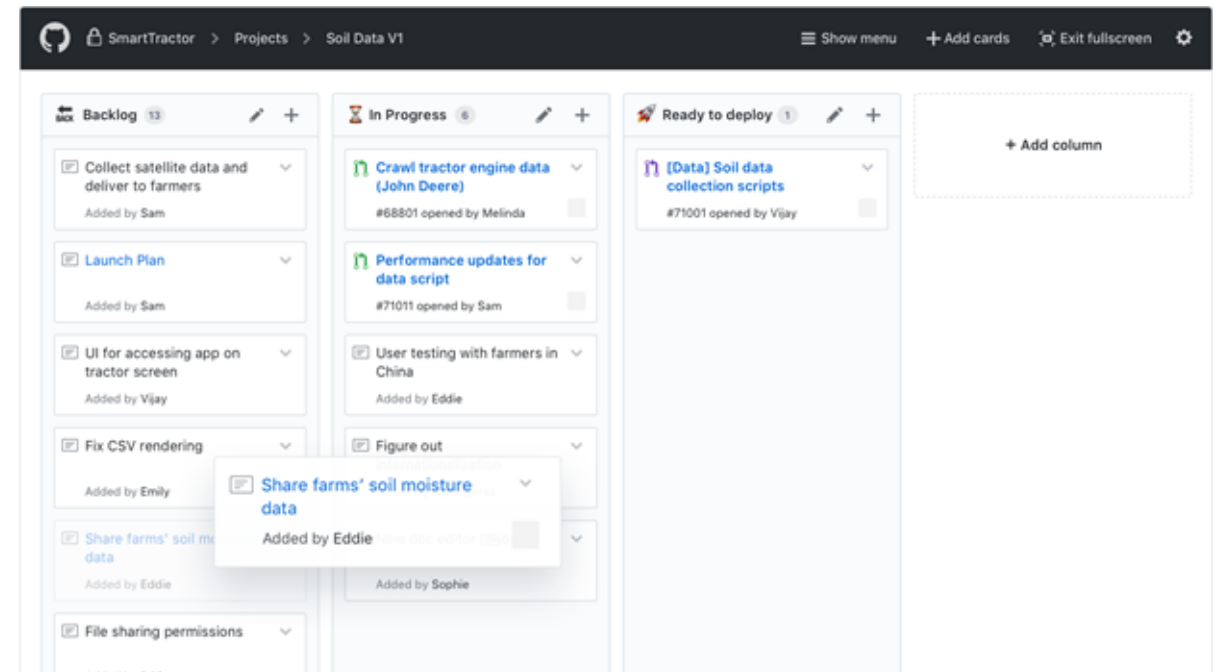
- Estimated

- Assigned to individual team members

- Acceptance criteria should be defined

User Stories are often used

Kanban boards



Scrum Meetings

Sprint Planning Meeting

Entire Team decides together what to tackle for that sprint

Daily Scrum Meeting

Quick Meeting to touch base on :

What have I done? What am I doing next? What am I stuck on/need help?

Sprint Retrospective

Review sprint process

Sprint Review Meeting

Review Product

Standups



User stories

- Plan using units of customer-visible functionality



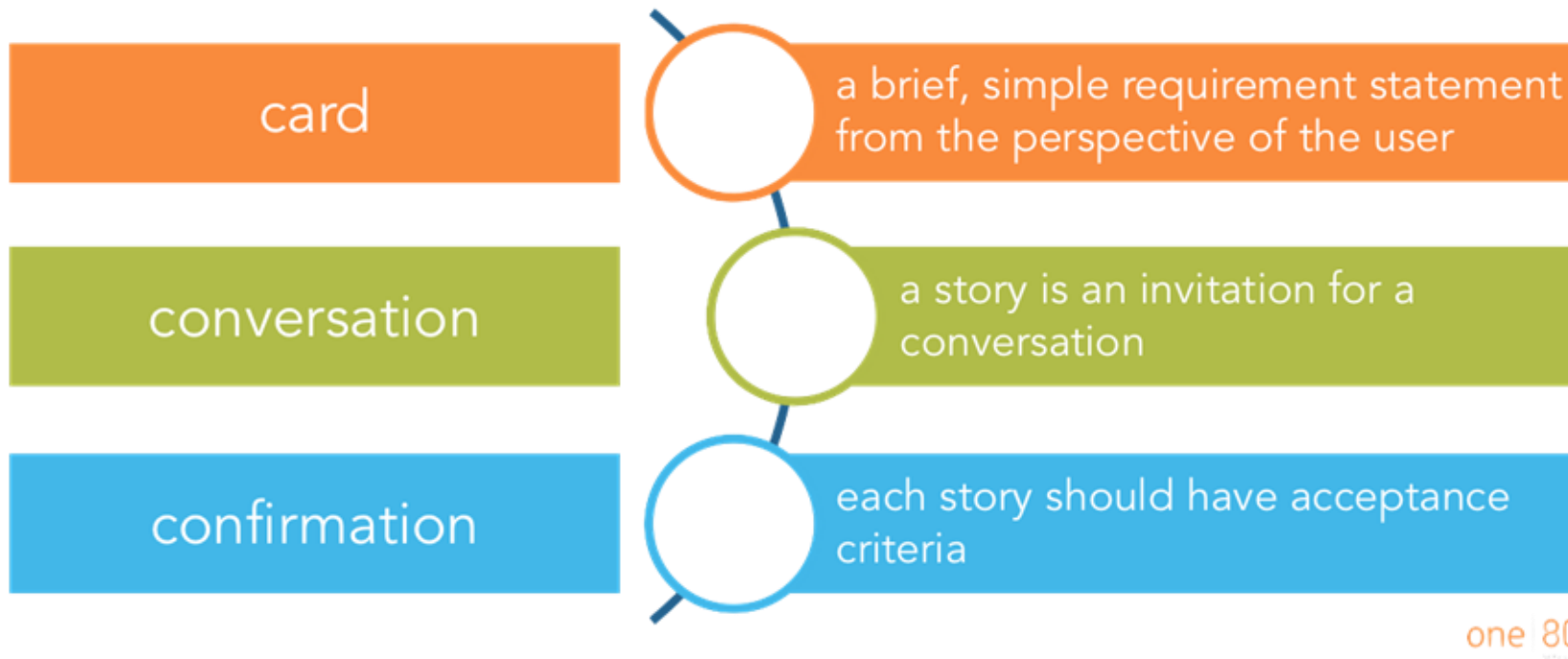
Example

Title: Order Flight DVD

Description: A user will be able to order a DVD of a flight they have been on.

.....

User Stories



User story cards (3"x5")

“As a [role], I want [function], so that [value]”

Conversation

- Developers, product managers, etc.
- Is it clear to everyone?
- What must a developer do to implement this user story?

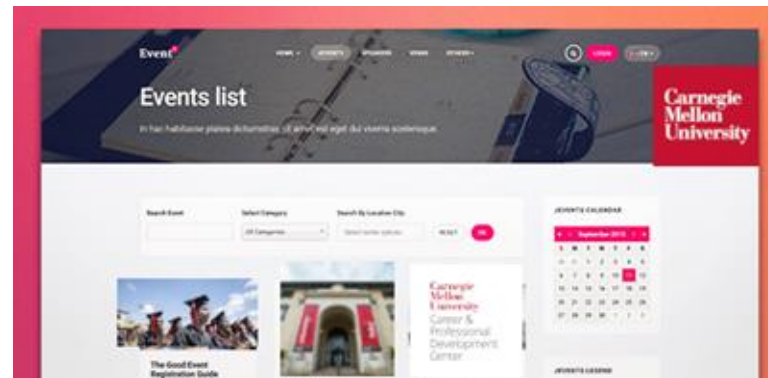
Acceptance criteria

- How can we tell that the user story has been achieved?
- It's easy to tell when the developer finished the code.
- But, how do you tell that the customer is happy?

Example

“As a [role], I want [function], so that [value]”

The university is looking to enhance student and staff engagement by creating an online platform where all university-related events are easily accessible. The goal is to provide a user-friendly website that serves as a central hub for information on various activities, ranging from academic seminars to sports events and club meetings.



Participation activity: write a user story

- Project to consider: university event website introduced on previous slide
- Write one story on the paper we give you
- Keep it and we'll revisit it later in the lecture

How to evaluate user story?

Follow the INVEST
guidelines for good
user stories!



I	independent
N	negotiable
V	valuable
E	estimable
S	small
T	testable

Source: <http://one80services.com/user-stories/writing-good-user-stories-hint-its-not-about-writing/>

one 80
services

Participation activity: write a user story

- Project to consider: university event website introduced on previous slide
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Independent



- Schedule in any order.
- Not always possible

Counterexample

As a student, **I want to** receive notifications for events that are about to start, for those I have shown interest in, **so I** don't miss them.

Acceptance Criteria:

- An option is provided to 'Set a Reminder' for each event.
- Notifications are sent to users who have opted for reminders, shortly before the event starts.

Assume that the homepage with an event calendar is already in place.

I	independent	☒
N	negotiable	☐
V	valuable	☐
E	estimable	☐
S	small	☐
T	testable	☐

Negotiable



- Details to be negotiated during development
- Good Story captures the essence, not the details

Counterexample

As a student, I want to view the upcoming events at the university, **so I** can decide which ones to attend.

Acceptance Criteria:

- Add an interactive grid layout of upcoming events at the top of the homepage.
- Each event card in the grid is visible for a 2 seconds before automatically rotating to display the next set of events.
- Each card in the grid includes the event's name, type (e.g., seminar, sports game), duration, a brief description, and scheduled times.
- This grid of events is displayed under a prominent H1 heading that reads "Discover What's Happening on Campus!"

I	independent	☒
N	negotiable	☒
V	valuable	☐
E	estimable	☐
S	small	☐
T	testable	☐

Valuable



- This story needs to have value to someone (hopefully the customer)
- Easy to forget *why* you are doing what you are doing

Counterexample

As the Events Coordinator, **I want** a database to store details of students and staff interested in university events.

Acceptance Criteria:

- A database is constructed to manage user information.
- The database stores details such as name, email, phone number, favorite event types, date of birth, and history of event attendance or registrations.

I	independent	☒
N	negotiable	☒
V	valuable	☒
E	estimable	☐
S	small	☐
T	testable	☐

Estimable



- Helps keep the size small
- It should provide enough details to estimate the amount of effort needed
- More on estimates later...

Counterexample

As an undergraduate student, **I want to** be able to filter university events, **so I** can choose the ones that align with my interests.

Acceptance Criteria:

- Filters are added to the event listings on the website.

I	independent	☒
N	negotiable	☒
V	valuable	☒
E	estimable	☒
S	small	☐
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Small



- Fit on 3x5 card
- At most two person-weeks of work (one sprint)
- Too big == unable to estimate

Counterexample

As a student, I want to easily find information about upcoming events, **so** I can participate in activities that interest me.

Acceptance criteria:

- A homepage is created displaying the university's name, motto, location, email, and contact information.
- The homepage features a calendar of upcoming university events.
- The event calendar includes details such as the event title, type (e.g., seminar, sports game, club meeting), a brief description, location, date, and time.
- Users can filter the event list by event type, date, and hosting department or club.
- The admin can update the event calendar as new events are planned or existing events are modified.

I	independent	☒
N	negotiable	☒
V	valuable	☒
E	estimable	☒
S	small	☒
T	testable	☐

Testable



- Ensures understanding of task
- We know when we can mark task “Done”
- Unable to test == do not understand

Counterexample

As a student, **I want to** easily view promotional videos or trailers of university events, **so I** can decide which events to attend.

Acceptance Criteria:

- Promotional videos can be embedded on each event detail page.
- Videos are of high quality.
- The embedded video is well-integrated into the page design.
- The video size is large enough to ensure clarity.
- The video controls are user-friendly.



Activity Part 2: Evaluate using INVEST

Follow the INVEST
guidelines for good
user stories!

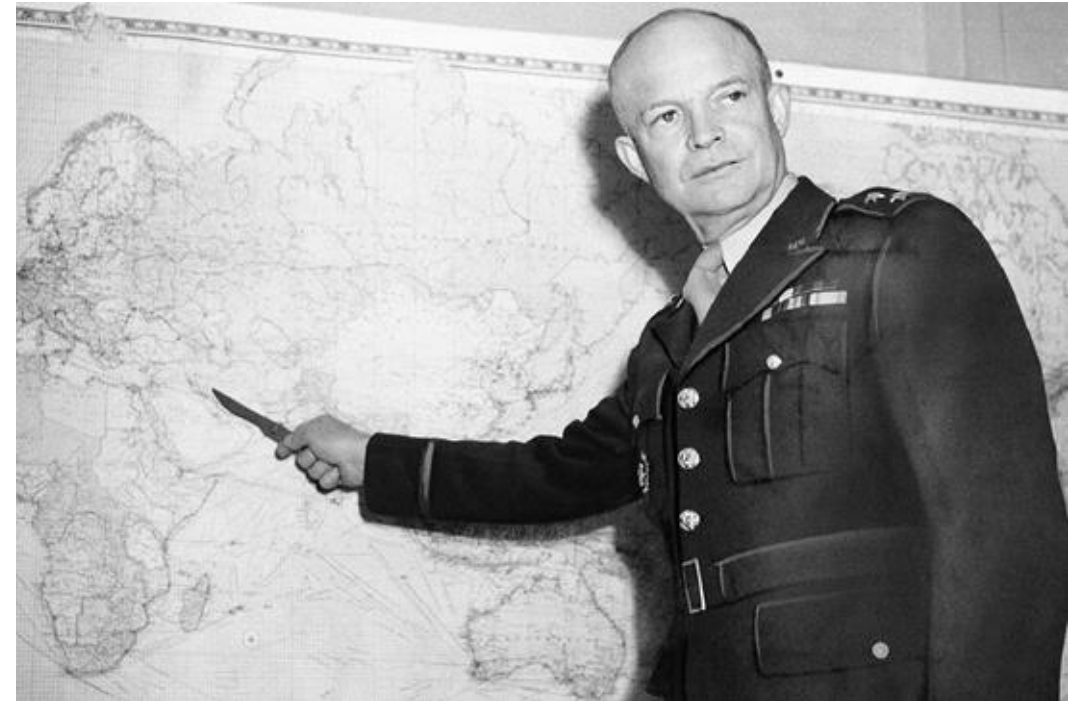


one | 80
SERVICES

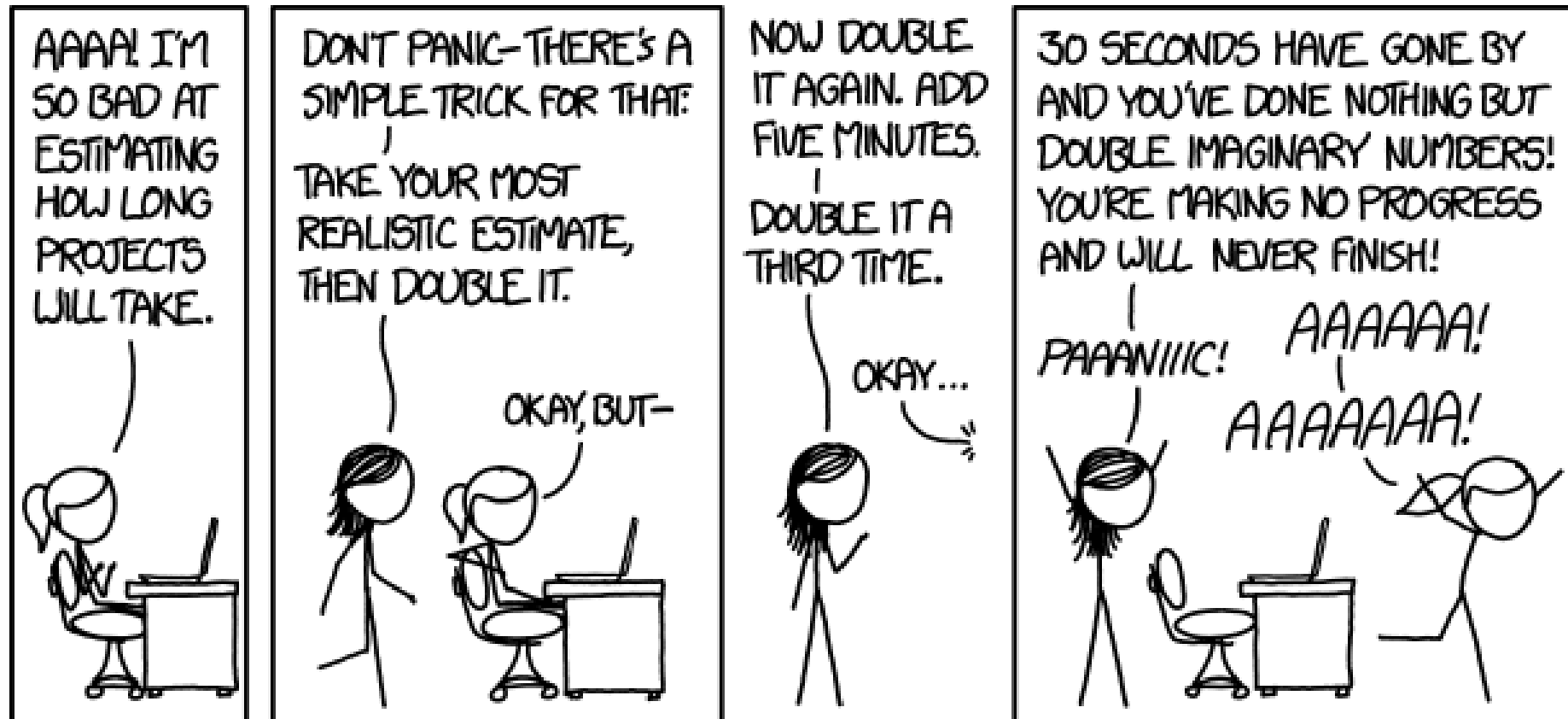


***“Plans are nothing,
planning is everything”***

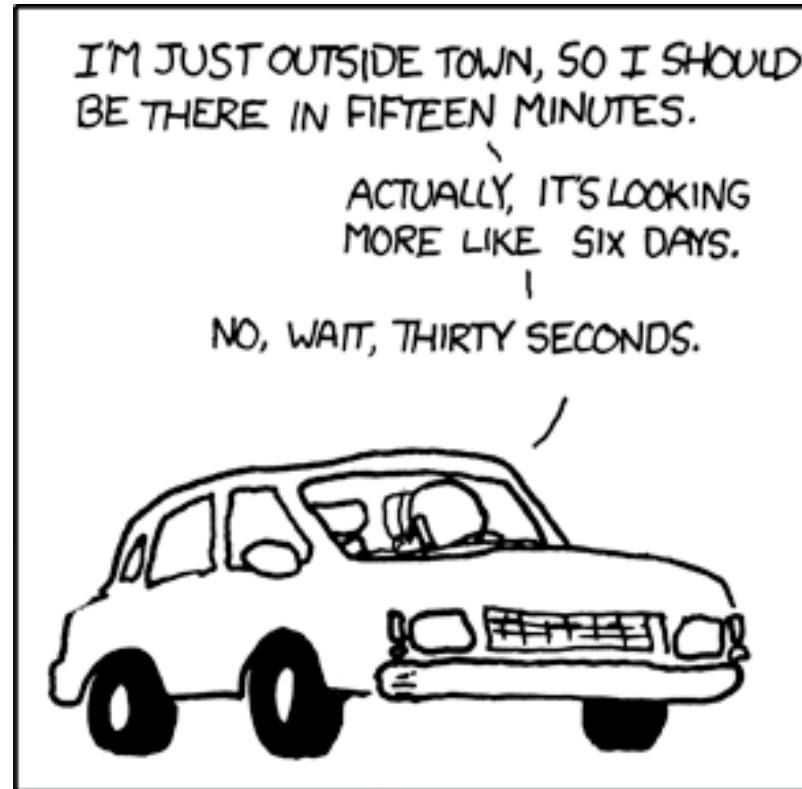
-Dwight D. Eisenhower



Time estimation



Time estimation



THE AUTHOR OF THE WINDOWS FILE
COPY DIALOG VISITS SOME FRIENDS.

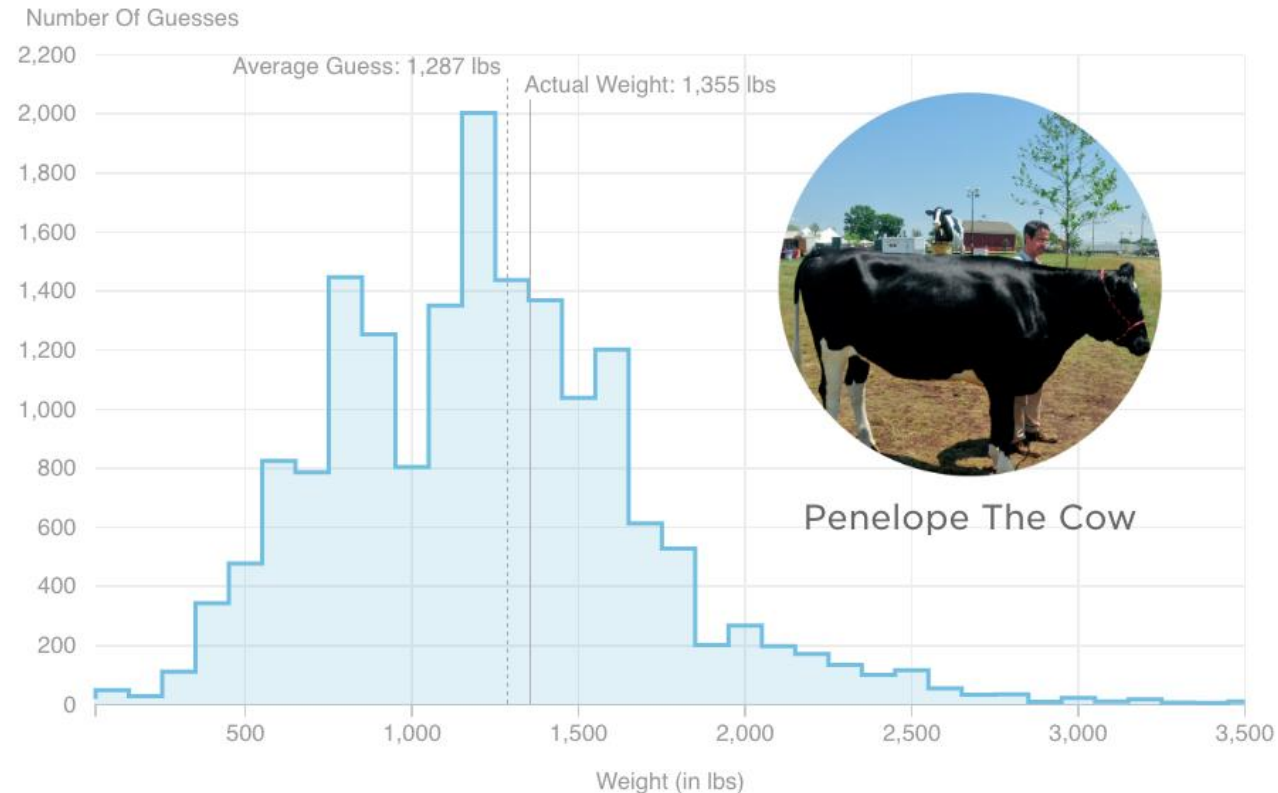
Improving Time Estimates

- Prevent conformity bias
- Do you have a comparable experience to base an estimate on?
- How much design do you need for each task?
- Break down the task into smaller tasks and estimate them.

Wisdom of the Crowd

How Much Does This Cow Weigh?

(All People)



66



XS



S



M



L



XL

made by **:codica**

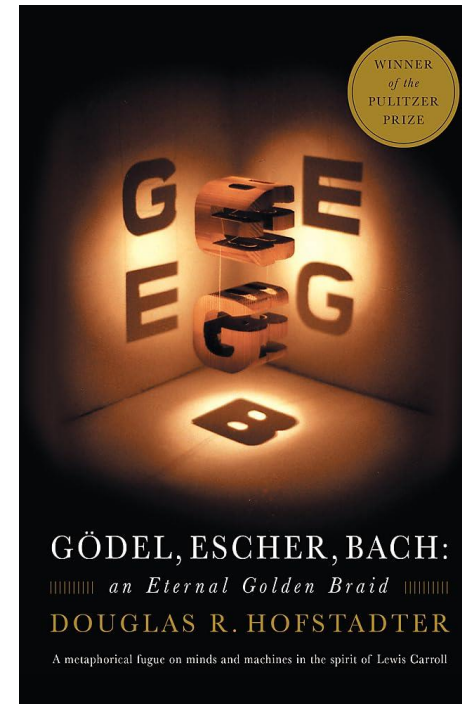
codica.com



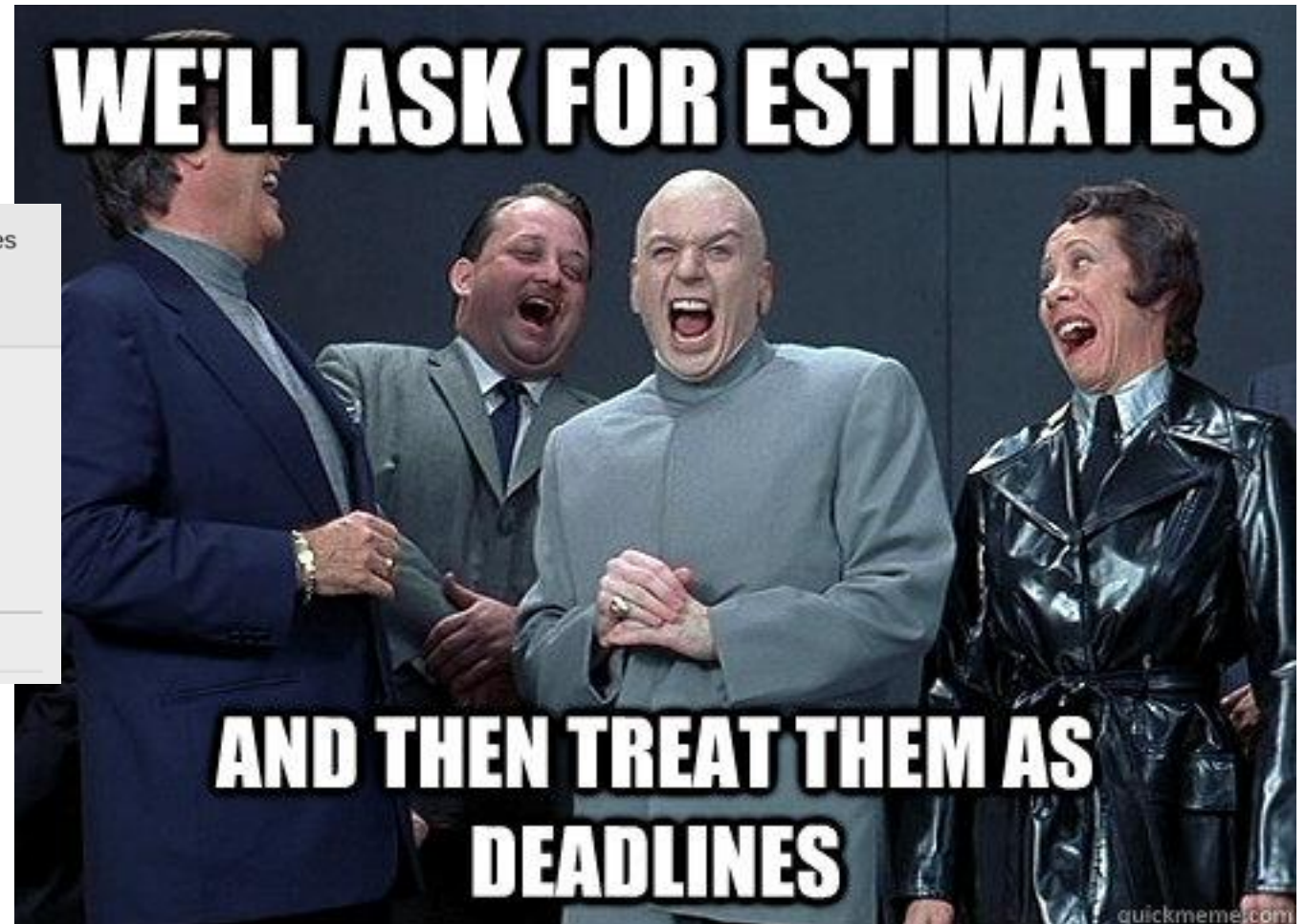
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Hofstadter's Law

"It always takes longer than you expect, even when you take into account Hofstadter's Law"



Is Estimation Evil?



Ron Jeffries

About Search Site Categories

Estimation is Evil

© Feb 1, 2013 • [Agile-Related, estimation]

The following article is recovered from the February 2013 issue of the Pragmatic Programmers magazine.

Overcoming the Estimation Obsession

Ron Jeffries's essay [Estimation is Evil](#)

Milestones and deliverables make progress *observable*

Milestone: clear end point of a (sub)tasks

- For project manager
- Reports, prototypes, completed subprojects
- "80% done" is not a suitable milestone

Deliverable: Result for customer

- Similar to milestones, but for customers
- Reports, prototypes, completed subsystems

What you need to know

- Recognize the importance of having a software process
- Main ideas of Agile/Scrum
- Understand backlogs and user stories
- Understand the difficulty of estimating tasks and progress
- We use milestones for planning and progress measurement

Reminders

- Problem Set 1 on Canvas, due September 2 @ 11:59pm
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